

Quantitative Evaluation of Line-Edge Roughness in Various FinFET Structures: Bayesian Neural Network With Automatic Model Selection

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ABSTRACT To design a device that is robust to process-induced random variation, this study proposes a machine-learning-based predictive model that can simulate the electrical characteristics of FinFETs with process-induced line-edge roughness. This model, i.e., a Bayesian neural network (BNN) model with horseshoe priors (Horseshoe-BNN), can significantly reduce the simulation time (as compared to the conventional technology computer-aided design (TCAD) simulation method) in a sufficiently accurate manner. Moreover, this model can perform autonomous model selection over the most compact layer size, which is necessary when the amount of data must be limited. The mean absolute percentage error for the mean and standard deviation of the drain-to-source current (I_{DS}) were $\sim 0.5\%$ and $\sim 6\%$, respectively. By estimating the distribution of the current-voltage characteristics, the distributions of the other device metrics, such as off-state leakage current and threshold voltage, can be estimated as well.

INDEX TERMS Line edge roughness (LER), process-induced random variation, Bayesian neural network, automatic model selection.

I. INTRODUCTION

As the lateral/vertical dimension of transistors in integrated circuits (ICs) (e.g., channel length, channel width, junction depth, etc.) has been scaled down, the process-induced random variation of device parameters becomes significant. This would significantly damage the yield of devices. Process-induced line-edge-roughness (LER) is one type of random variation sources, and it induces a significant amount of variation in aggressively scaled transistors. Because the amplitude of LER did not shrink as much as the feature size shrinkage, the portion of LER in the nominal/physical channel length and width became significant, resulting in a larger variation in $I_{DS} - V_{GS}$ characteristics of devices. Hence, the LER in device has become a primary limit when the physical dimensions of device has been scaled down. In this regard, precise profiling of the impact of LER on the device performance has become

an indispensable requisite for designing a variation-immune device. To date, technology computer-aided design (TCAD) simulation has been used to evaluate the impact of LER on device characteristics; however, it takes a significant amount of time. Thousands of devices should be simulated to determine the exact amount of LER-induced variation (in real, a few weeks even for a single LER profile). For this reason, a novel approach for less time-consuming simulation is necessary. In this context, we proposed a linear regression model in [1]; however, this linear regression model was limited in dealing with nonlinearity. Conversely, artificial neural network (ANN) is the most popular machine-learning model because it shows a powerful ability to find patterns as well as to detect trends in complex real-world data. By stacking multiple layers (in detail, neurons/nodes aggregations that receive inputs and compute/give outputs, based on their pre-defined simple non-linear functions), an ANN can well define highly nonlinear functions and capture non-linear dependencies in data. However, standard ANN models are prone

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to overfitting and exhibit poor generalization performance. In addition, they tend to make unjustified over-confident predictions for the inputs far away from the training data, which can sometimes result in suboptimal predictions [2]. In particular, in the regime of small data (where the uncertainty of data is severe), ANNs are prone to poorly specify their weights. To solve these problems, the Bayesian neural network (BNN) has received lots of attention. The BNNs treat their weights as random variables (described by distributions), and thereby capturing uncertainties in the training data and their weights. This study employed BNN models to predict the $I_{DS} - V_{GS}$ characteristics in various FinFETs with different device structures. In real, there is a demand for FinFET devices with multiple heights [3], [4]. The BNN model proposed in this work assigns horseshoe priors [5] as a prior distribution, which we call the Horseshoe-BNN (HS-BNN). The prior (i.e., prior distribution) is a prior belief of the weight distribution's configuration before observing the data. According to the Bayesian theorem, the beliefs are updated after the data observance [6]. Herein, depending on which prior distribution is to be specified, it leads to a different model training process. While the Gaussian prior is the most commonly-used prior, the horseshoe distribution has been chosen to introduce sparsity and shrinkage over the weights. The horseshoe prior makes it available the model selection over a number of nodes and can reduce arduous/tiresome work in optimizing the layer sizes.

II. SIMULATION METHODOLOGY AND DATA PREPARATION

Various FinFETs with different structures were designed on the basis of [7]–[10], and thereafter, 3D LER profiles on those FinFETs were applied/simulated using the MATLAB model (which was proposed in [11]) and Sentaurus TCAD. Specifically, the 3D LER sequences (i.e., points of the randomly rough surfaces) were generated by the MATLAB model with three main parameters for LER [i.e., RMS amplitude (σ), $x(y)$ -axis correlation length ($\xi_X(\xi_Y)$), and roughness exponent (α)], and the sequences were imported into the TCAD tool. Herein, the RMS amplitude (σ) indicates the standard deviation of the LER amplitudes, and it is often referred to as the LER or LER magnitude. Note that σ is the major factor for LER. The correlation length (ξ) is corresponding to the wavelength of LER profile, and the roughness exponent (α) quantitatively indicates the way how high-frequency components in LER profile diminishes. The device parameters for FinFET (including the parameters for LER and device structure) are summarized in Table 1. Figure 1(a) shows an isometric view and current density distribution of a FinFET with $L_g = 20$ nm, $W_{fin} = 7$ nm, $H_{fin} = 42$ nm. Note that the feature parameters for LER used in the FinFET is as follows: $\sigma = 0.5$ nm, $\xi_X = 20$ nm, $\xi_Y = 50$ nm, $\alpha = 1$. Figure 1(b) shows the $I_{DS} - V_{GS}$ characteristics with 100 sample devices. Herein, the fin width variation by LER causes the variations in the $I_{DS} - V_{GS}$ characteristics [12]. It is noteworthy that the 3D TCAD device simulations were run using various

TABLE 1. Device parameters used in this work.

Device Parameters	Value
Gate length (L_g)	14–26 nm
Fin width (W_{fin})	6–8 nm
Fin height (H_{fin})	24–53 nm
Gate oxide thickness (T_{ox})	3 nm
Substrate doping concentration (N_{sub})	10^{15} cm $^{-3}$
Source/drain doping concentration ($N_{S/D}$)	10^{20} cm $^{-3}$
Peak concentration of punch-through stopper	2×10^{19} cm $^{-3}$
Power supply voltage (V_{DD})	0.7 V
RMS amplitude (σ)	0.1–1.3 nm
Correlation length along x direction (ξ_X)	0.5–200 nm
Correlation length along y direction (ξ_Y)	1–500 nm
Roughness exponent (α)	0.1–1.3

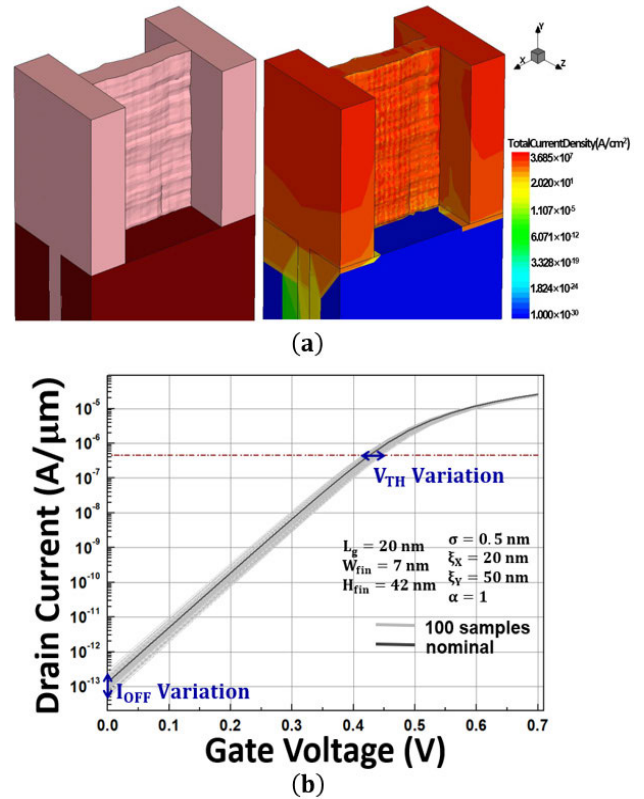


FIGURE 1. (a) Isometric view of FinFET with LER (left) and its current density distribution (right), and (b) simulated $I_{DS} - V_{GS}$ of nominal device and 100 sample devices with LER. The parameters for LER and FinFET device structure is $\sigma = 0.5$ nm, $\xi_X = 20$ nm, $\xi_Y = 50$ nm, $\alpha = 1$, and $L_g = 20$ nm, $W_{fin} = 7$ nm, $H_{fin} = 42$ nm, respectively.

physics models, i.e., the Shockley–Read–Hall model for carrier generation and recombination, the Old Slotboom model for bandgap narrowing, the Lombardi model for thin-layer mobility, and the density gradient quantization model for quantum mechanics effects.

We have noted that creating a number of data is very time-consuming because of a long TCAD simulation run

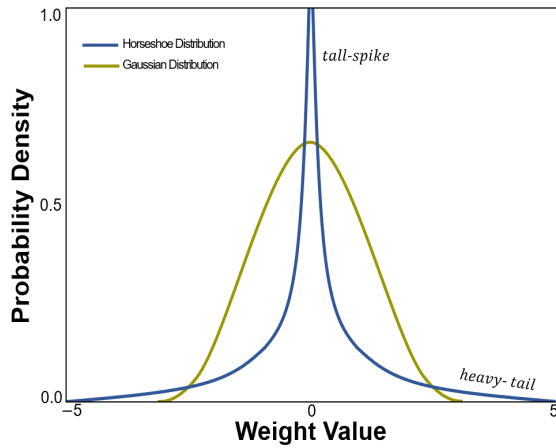


FIGURE 2. Probability density of Horseshoe prior and Gaussian prior.

time. This led to a dilemma when satisfying (i) the number of LER profiles or (ii) the number of sample devices with each LER profile. In other words, to appropriately estimate the distribution of electrical characteristics, the number of sample devices should be sufficient, but at the same time, the number of LER profiles is quite important. In considering those two aspects, 169 different datasets were composed for training and testing the model, and each dataset consisted of 50 different sample FinFETs with identical LER and device parameters. Eighteen types of FinFET device structure were chosen based on [7]–[10], and the LER parameters were randomly selected in the relevant ranges shown in Table 1.

As mentioned in the previous study [1], the distribution of $\log(I_{DS})$ for a given V_{GS} can be considered as a normal distribution in terms of small kurtosis and skewness values. From this perspective, the mean and standard deviation of $\log(I_{DS})$ (denoted as μ_{IDS} and σ_{IDS} , respectively) were selected as target variables. And gate voltage (V_{GS}), LER parameters (i.e., RMS amplitude (σ), correlation length along the x-axis (ξ_X), correlation length along the y-axis (ξ_Y), and roughness exponent (α)) and device structure parameters (i.e., gate length (L_g), fin width (W_{fin}), and fin height (H_{fin})) were selected as feature variables to predict the LER-induced variation in the I_{DS} - V_{GS} characteristics of various FinFET structures.

III. BAYESIAN NEURAL NETWORK WITH AUTOMATIC OPTIMIZATION ON LAYER SIZE

Predictive models are just a function in the form of $y = f(x)$ that fits the given data $D = \{(x_n, y_n)\}_1^N$, and they are generalized/extended to the cases that have not been seen in training. For example, the function form of a linear regression model is $y = \sum w_i x_i + b$. However, this model has its own limit for determining the nonlinear relationship between x and y . Conversely, an ANN with an L hidden layer is in the form of $y = W^{(L+1)}h^{(L)} + b^{(L+1)}$, where W^l is a matrix of weights between layer $(l-1)$ and layer (l) in the size of $\mathbb{R}^{(K_{l-1}+1) \times K_l}$, and $h^{(l)}$ is a matrix of

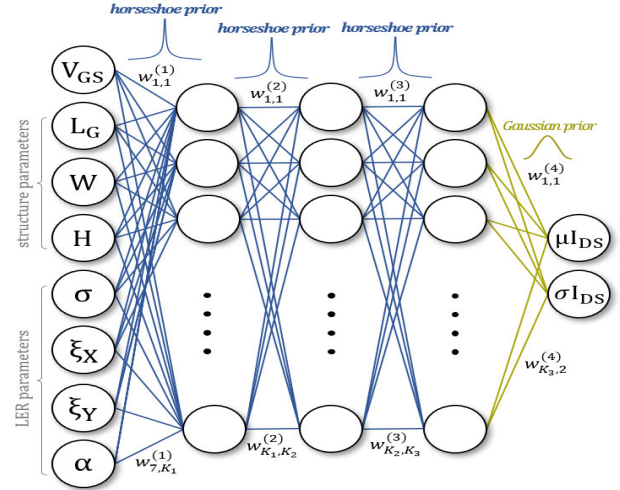


FIGURE 3. Architecture of the HS-BNN, predicting the mean and standard deviation of $\log(I_{DS})$ in various FinFET structures with arbitrary LER profiles.

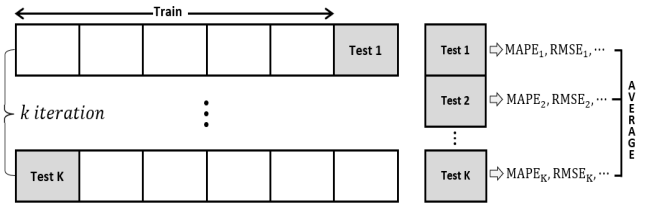


FIGURE 4. Diagram of K-fold cross-validation.

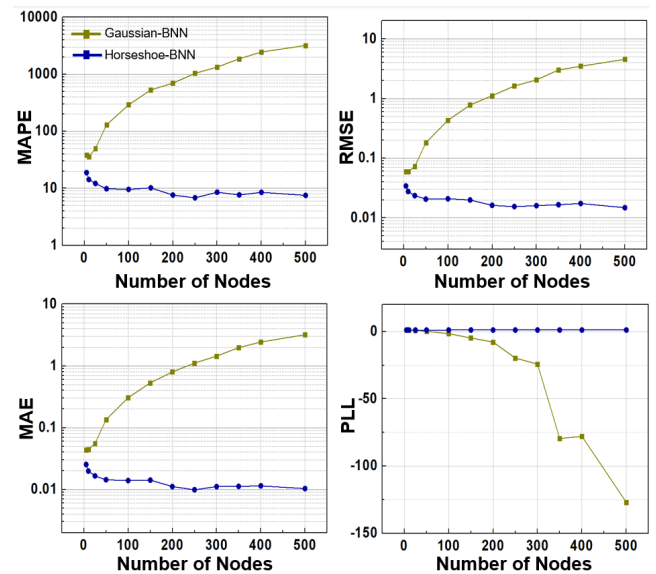


FIGURE 5. MAPE, RMSE, MAE, and PLL values of each model along different numbers of nodes, showing model selection over the number of HS-BNN nodes.

outputs from layer (l) . Herein, K_l is the number of units in layer l , $h^{(l)} = a(W^{(l)}h^{(l-1)} + b)$, $h^{(0)} = x$, and $a(\cdot)$ is an activation function for each hidden layer. In the case of the ANN with three hidden layers, the function is

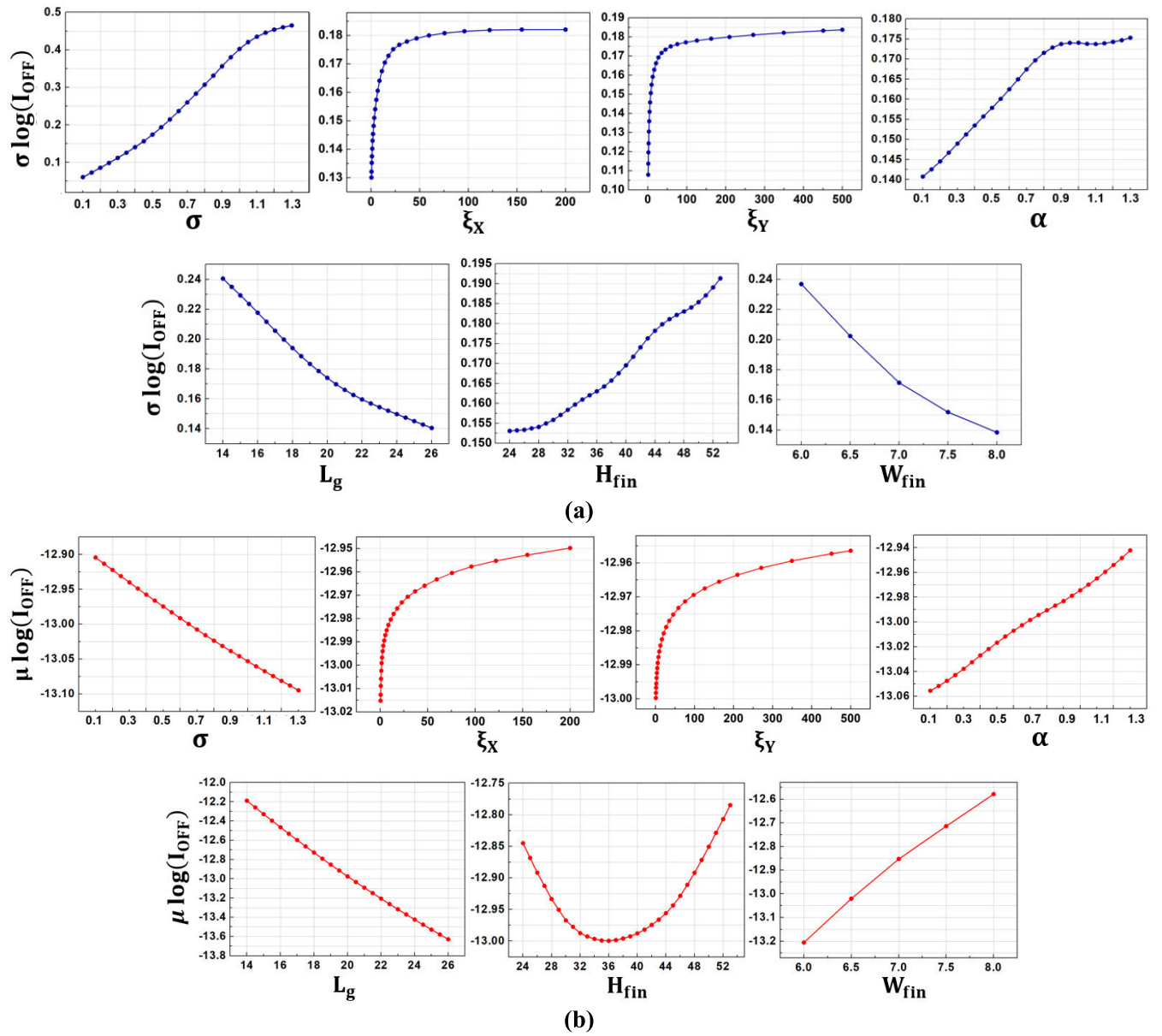


FIGURE 6. (a) Predicted standard deviation and (b) predicted mean of $\log(I_{OFF})$ for FinFET structure of $L_g = 20$ nm, $H_{fin} = 42$ nm, $W_{fin} = 7$ nm with the LER profile of $\sigma = 0.5$ nm, $\xi_X = 20$ nm, $\xi_Y = 50$ nm, $\alpha = 1$. Note that each parameter (i.e., σ , ξ_X , ξ_Y , α , L_g , H_{fin} , and W_{fin}) is varied in relevant ranges.

$y = W^4 a(W^3 a(W^2 a(W^1 x + b_1) + b_2) + b_3) + b_4$. The ANN model is a composite function of multiple activation functions (i.e., activation integral) [13], which can make it complex nonlinear regression predictions.

A Bayesian neural network is an extended standard ANN with Bayesian/posterior inference [14]. While the ANN is a deterministic model, the BNN is a stochastic/probabilistic model. The BNN incorporates uncertainty in modeling tasks by introducing distributions over the weights $W \sim p(W)$ (i.e., $W = \{W^l\}_1^{L+1}$, a set of weight matrices over all the layers) [15]. By doing so, BNNs can demonstrate/reconfigure epistemic uncertainty when the amount of data is limited. However, at the same time, there are inherent neural network problems like over

(under)-parameterization. The larger(smaller) the number of data points is, the larger(smaller) the number of required nodes is. When the number of nodes is much larger(smaller) than that estimated from the data, the variance around the BNN's predictions would be too large(small). The layer sizes should be as large as necessary. Especially, in the case of estimating the LER-induced variation, data were bound to be small, so that over-parameterization had to be prevented. In this regard, this study assigns horseshoe priors over weights using Bayesian inference. The horseshoe prior is given by: $(w_{i,j}^{(l)} | \tau_i, \nu) \sim N(0, \tau_i^2 \nu^2)$ where $\tau_i \sim C^+(0, b_0)$ and $\nu \sim C^+(0, b_g)$. Herein, $w_{i,j}^{(l)}$ is the weight connecting the i^{th} node of layer $(l-1)$ and j^{th} node of

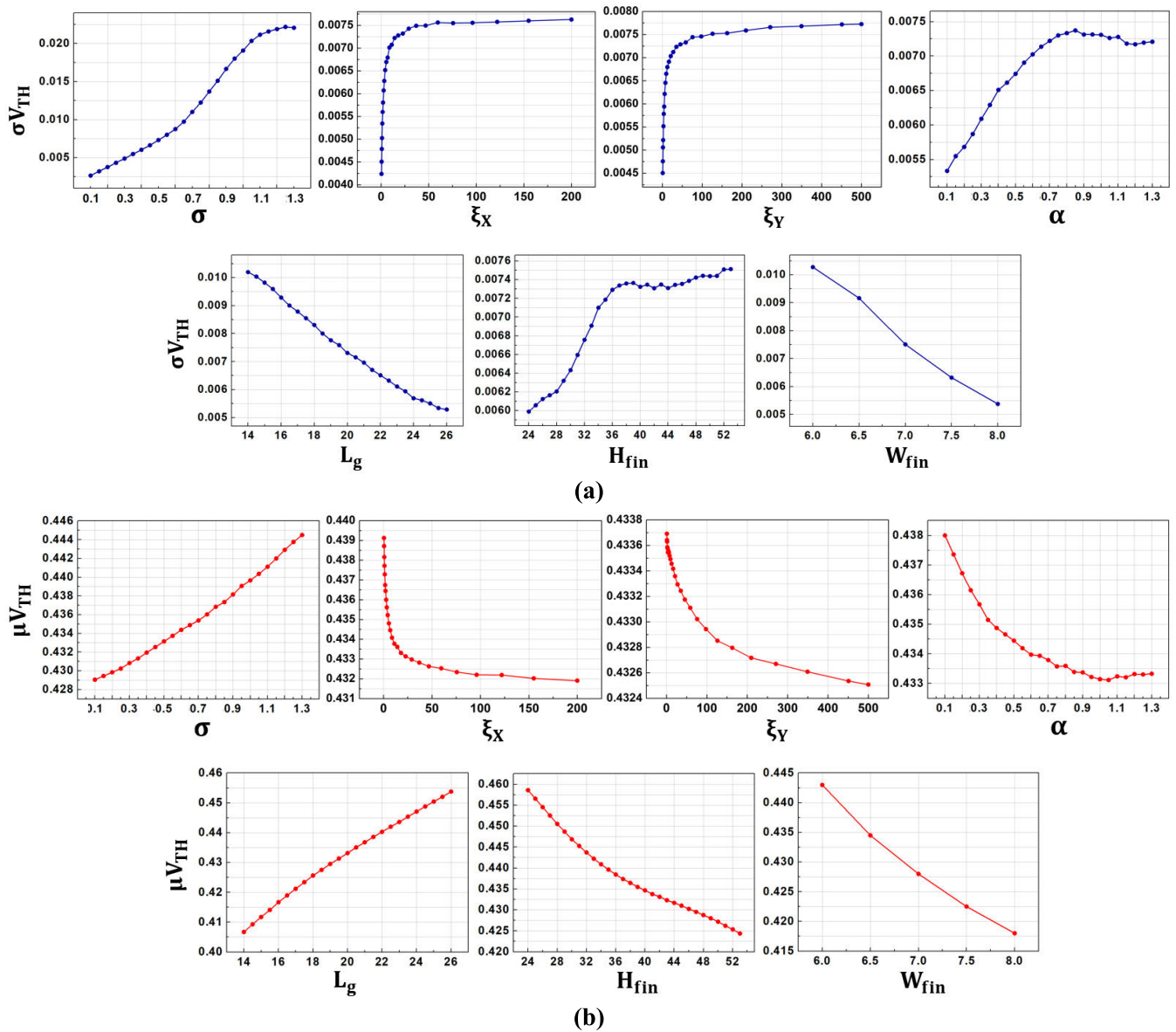


FIGURE 7. (a) Predicted standard deviation and (b) mean of V_{TH} distribution for FinFET structure of $L_g = 20$ nm, $H_{fin} = 42$ nm, $W_{fin} = 7$ nm with the LER profile of $\sigma = 0.5$ nm, $\xi_X = 20$ nm, $\xi_Y = 50$ nm, $\alpha = 1$. Note that each parameter (i.e., σ , ξ_X , ξ_Y , α , L_g , H_{fin} , and W_{fin}) is varied in relevant ranges.

layer (I), and $C^+(\mathbf{0}, \mathbf{b})$ is a half-Cauchy distribution, and $\mathbf{b}_0$ and $\mathbf{b}_g$ are shrinkage parameters. The horseshoe prior can introduce shrinkage and sparsity over the weights and bring model selection that automatically finds the most compact neural network structure (i.e., finding the best number of nodes). This is due to the two features of the horseshoe distribution: (1) a tall spike at zero and (2) heavy (relatively flat) tails (see Figure 2). The tall spike at zero promotes shrinkage of the weights, which seems unnecessary in making predictions. However, its heavy tail allows large weights to avoid shrinkage.

The complete architecture of the HS-BNN is illustrated in Figure 3: Three hidden layers with a RELU activation function, horseshoe priors for the weights into the hidden

layers, and a Gaussian prior for the weights into the output layer. The Gaussian prior was used to prevent overfitting [16].

IV. RESULTS AND DISCUSSIONS

The HS-BNN predictions were evaluated using K-fold cross-validation [splitting data in a K number of sections, and then, iteratively using one of the sections as a test set and the others as a training set (see Figure 4)]. This method is expected to be a proper choice when data are limited, by ensuring all data to be used as a test set at least once, which makes less biased evaluation and well estimates how a model will perform in general cases [17], [18]. Mean absolute percentage errors (MAPEs), root-mean-squared errors (RMSEs), mean absolute errors (MAEs) (herein, note that the lower they are,

TABLE 2. MAPE and RMSE values of each model.

Model	Target	MAPE	RMSE
HS-BNN	μI_{DS}	0.540%	0.0536
	σI_{DS}	6.659%	0.0101
BLR	μI_{DS}	0.805%	0.0804
	σI_{DS}	19.586%	0.0237

TABLE 3. MAPE, RMSE, MAE, and PLL values of each model.

Model	Target	MAPE	RMSE	MAE	PLL
HS-BNN	μI_{DS}	0.546%	0.071	0.052	-1.598
	σI_{DS}	6.800%	0.015	0.009	1.178
Gaussian-BNN	μI_{DS}	6.658%	0.794	0.602	-1.683
	σI_{DS}	35.671%	0.059	0.044	1.070

the better they are), and predictive log likelihoods (PLLs) (herein, note that the higher it is, the better it is) are calculated in each test set. Afterwards, the averages of those are the representative evaluation process values.

The overall prediction performance was improved, as compared against the previous work [1] which is the Bayesian linear regression (BLR) model (see Table 2). Note that the HS-BNN model for this comparison is only for the FinFET with $L_g = 14$ nm, $W_{fin} = 7$ nm, $H_{fin} = 30$ nm because the BLR model made predictions only for the corresponding FinFET structure. Table 2 summarizes the MAPE and RMSE of the two target variables (i.e., μI_{DS} and σI_{DS}) of the HS-BNN and BLR. The predictions of both μI_{DS} and σI_{DS} were much improved over those in the previous work. The prediction for μI_{DS} was improved (i.e., 0.55% vs. 0.81%), but there was significant prediction improvement for σI_{DS} (i.e., 6.66% vs. 19.59%).

Table 3 and Figure 5 show the results of HS-BNN model for various FinFET structures. To verify the benefit of the HS-BNN model, we compared it with the Gaussian-BNN model. Table 3 summarizes the predictive performance, showing that the HS-BNN showed much better results. Figure 5 demonstrates the performance of the model selection with respect to the layer sizes. For the number of nodes over 200, the HS-BNN's prediction for σI_{DS} showed almost the same results (MAPEs $\sim 7\%$), while the Gaussian-BNN showed totally different results for different numbers of nodes. It is noteworthy that it is a log-scale for the MAPE, RMSE, and MAE in Figure 5. This indicates that, as mentioned in [19], even if the number of nodes was excessively overestimated, the horseshoe prior made it possible to find the most compact layer sizes. However, it is meaningful, not to merely make a simple comparison of predictive performance, but to focus on the fact that the HS-BNN achieved the model selection while making a fine predictive performance. This is because the number of nodes for every hidden layer is specified as the same. Manual layer size optimization was not performed, and the optimization was performed automatically

in the HS-BNN. Thus, it is an open question whether the HS-BNN is better than all the Gaussian-BNNs. However, it is certain that the HS-BNN has saved a significant optimization time.

Figures 6 and 7 show the predictions of the mean and standard deviation of log off-state current [$\log(I_{OFF})$] and threshold voltage (V_{TH}) of the FinFET with $L_g = 20$ nm, $W_{fin} = 7$ nm, $H_{fin} = 42$ nm, and an LER profile [i.e., $\sigma = 0.5$ nm, $\xi_X = 20$ nm, $\xi_Y = 50$ nm, $\alpha = 1$]. By fixing all the parameters other than a single parameter, $\sigma \log(I_{OFF})$, $\mu \log(I_{OFF})$, σV_{TH} , and μV_{TH} were predicted.

The device performance trends shown in Figures 6 and 7 seem to be well matched to the general trends, when considering several studies shown in [20]–[25]. Using this HS-BNN model, we can show how both the standard deviation and mean of the electrical characteristics vary with arbitrary LER profiles in various FinFET structures.

V. CONCLUSION

A Bayesian neural network model with horseshoe priors (HS-BNN) has been proposed for the design of LER-immune FinFETs. This model can quantitatively estimate the LER-induced random variation of I_{DS} - V_{GS} characteristics in various FinFET structures within a few seconds, which is much shorter than the conventional TCAD simulation running time. With the capability of neural networks to define highly nonlinear functions, the prediction has been improved [vs. the Bayesian linear regression model [1] (i.e., $\sim 6\%$ vs. $\sim 19\%$)]. Moreover, the model selection performance over the most compact layer size was verified, which is essential when the amount of data is very limited.

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